🧊 3D Coordinates Extension — English Worksheet (Advanced)

Topic: Design Your Own 3D Build (x, y, z) · **Course:** 3D Coordinates · **Level:** Extension (Advanced, after Lesson 3) · **Time:** about 50 minutes

Pixel art was flat — two numbers, one wall. This lesson your build is **solid**. Three numbers. You draw it from the **front** and from **above**, then read both drawings back into the world.

Build your canvas. Run this to lay a blank **15 × 15** floor, then put a **red block at your feet** as your **home spot**:

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
        pos(1, 0, 1),
        pos(15, 0, 15),
        FillOperation.REPLACE)
player.on_chat("run", on_run)
```

Your build stands on that floor. **x** goes across, **y** goes up, **z** goes away from you — a **15 × 15 × 15** space.

> 🔴 **Stand on the red block every time you run.** If you move, the build moves too.

1 · Predict 🔮

Read each set of steps. Write what you will see **and the reason**.

```
place blue block at (3, 9, 4)
place blue block at (4, 9, 4)
place blue block at (5, 9, 4)
```

Three blocks share y = 9 and z = 4. Which way does the line run, which number changes, and why?


```
place blue block at (7, 2, 4)
place blue block at (7, 3, 4)
place blue block at (7, 4, 4)
place blue block at (7, 5, 4)
```

These four share x = 7 and z = 4. Which way does the line run, and why?

```
place blue block at (7, 5, 1)
place blue block at (7, 5, 2)
place blue block at (7, 5, 3)
```

These three share x = 7 and y = 5. Which way does the line run, and why?

```
place green block at (5, 6, 5)
place green block at (6, 6, 5)
place green block at (5, 6, 6)
place green block at (6, 6, 6)
```

These four share y = 6. What shape do they make, and would you see it from the front or from above?

```
place green block at (6, 6, 6)
place blue block at (6, 6, 6)
```

Two blocks aim for the same (x, y, z). What ends up in the world, and why?

2 · Spot the Bug 🐛

Each one is broken. Fix it, then explain why the original was wrong and why your fix works.

Bug A — The table's two legs should sit at (2, 1, 2) and (2, 1, 4). Both land in the **same spot**.

```
place brown block at (2, 1, 2)
place brown block at (2, 1, 2)
```

Write the fixed line:

Why was it wrong? Why does your fix work?

Bug B — This should be a line running **away** from you at x = 8, y = 5, from z = 3 to z = 5.

```
place black block at (8, 5, 3)
place black block at (9, 5, 3)
place black block at (10, 5, 3)
```

Write the fixed lines:

Why was it wrong? Why does your fix work?

Bug C — This should be a 2 × 2 × 2 cube with its bottom-left-near corner at (4, 4, 4). Eight blocks, eight different spots. It only makes a flat square.

```
place blue block at (4, 4, 4)
place blue block at (5, 4, 4)
place blue block at (4, 5, 4)
place blue block at (5, 5, 4)
```

Write the four missing lines:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📸 🎥

Stand on your **home spot** (the red block).

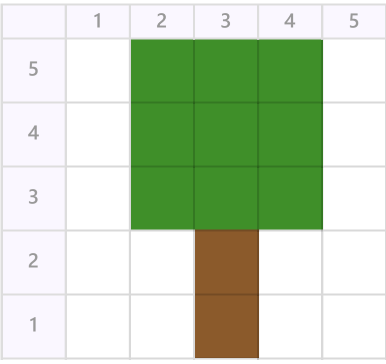
Your five rules.

#	Your design must…
1	fit inside the **15 × 15 × 15** space
2	use **3 or more** different block colours
3	use **40 or more** blocks
4	be **recognisable** — someone else can name it without being told
5	be **drawn on both grids below before you place a single block**

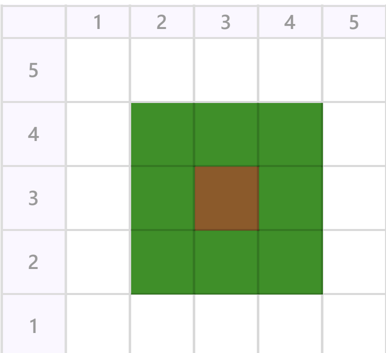
Two drawings, one build. The front view gives you **x** and **y**. The top view gives you **x** and **z**. Together they place every block.

Here is one I made — a tree:

🌳 Front view — x across, y up



🌳 Top view — x across, z away from you



Read it back. The trunk is at x = 3, z = 3, and stands from y = 1 to y = 2. The leaves fill x = 2 to 4 and z = 2 to 4, at y = 3 to y = 5.

Now draw yours. Colour the squares.

✏️ Your front view — x across, y up

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
15															
14															
13															
12															
11															
10															
9															
8															
7															
6															
5															
4															
3															
2															
1															

✏️ Your top view — x across, z away from you

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
15															
14															
13															
12															
11															
10															
9															
8															
7															
6															
5															
4															
3															
2															
1															

Write the plan. List the (x, y, z) of every block in your **biggest** colour, in the order you will place them.

Then build it. Read the front view for **x** and **y**. Read the top view for **x** and **z**. Place each block at its (x, y, z).

Check your work. Walk around your build. Look at it from the front, then from above. Write down every block you had to move, and what was wrong with your first (x, y, z).

Someone else names it. Show your build to a person who has not seen your grids. What did they say it was? If they got it wrong, what would you change?

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your grids. Hold this paper up to the camera.

2 · Your build. Point the camera. Hold your paper next to the build so both are in the shot.

3 · Your five rules. Check each one out loud against your build.

Fill the blanks:

> Today I built ______.
>
> I built it using this code: ______.

> In this code I used ______.
>
> The hardest part was ______.
>
> That part was hard because ______.
>
> The most fun part was ______.
>
> Something new I learned was ______.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.